

Discrete Mathematics Spring 2026

Homework 5: Sets

Due Sunday March 1 @11:59:00pm Eastern

Show all your work to receive full credit

1. (10 points) **True/False**

Which of the following are true? No justification needed, just write true or false.

- | | |
|--|---|
| 1. $\{2, \star\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ | 5. $\{2\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ |
| 2. $\{\{2, \star\}\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ | 6. $\{\star\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ |
| 3. $\{\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ | 7. $\{\star\} \in \{2, \{\star\}, \{2, \star\}\}$ |
| 4. $\star \in \{2, \{\star\}, \{2, \star\}\}$ | |

2. (15 points) Let A and B be two sets defined as follows:

- A is a set of size 6
- $B = \{1, 2, 3\}$

- List all elements of $P(B)$ (where $P(B)$ denotes the powerset of B).
- Calculate $|P(A)|$. Justify your answer.
- What is the maximum possible size for $|P(A \cup B)|$? What is the minimum? What must hold for each of those situations to be attained?

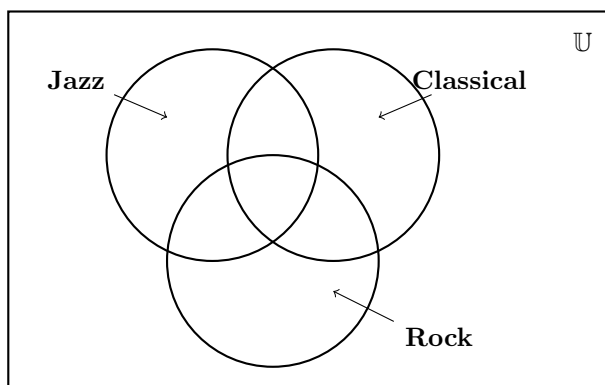
3. (15 points) Express each set in *set builder* notation. You may use connectives as well as existential and universal quantifiers. No justification needed.

Please do not use the predicates multiple, divisible, nor the operator divides $|$.

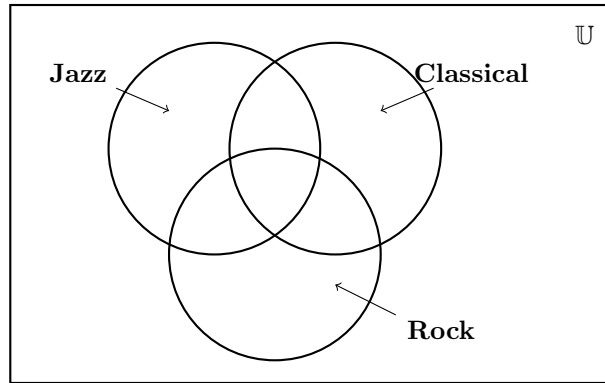
- $A = \{1, 4, 7, 10, 13, \dots\}$
- All negative integers greater than -300 that are **not** multiples of 5.

4. (10 points) For each part, shade the Venn diagram for the given set expression. The shaded region should represent exactly the given expression — partial credit will not be awarded for partially correct diagrams.

- Jazz** $\cup$ (**Classical** $\cap$ **Rock**)



(b) $\text{Jazz} \cap (\text{Classical} \cup \text{Rock})$



5. (15 points) Let A , and B be two sets.

We want to prove:

$$(A - B) \cap B = \{\}$$

Consider the following proof, where the steps were scrambled. Please put them in order. Write the number 1 next to the first step, 2 next to the second, etc. If a step has no place in the proof, please give it a zero. Note that at most one step can be given 0.

No need to give the laws used at each step.

[] $(A - B) \cap B = \{x | x \in A \wedge (x \notin B \wedge x \in B)\}$

[] $(A - B) \cap B = \{x | x \in A \wedge \text{False}\}$

[] $(A - B) \cap B = \{x | \text{False}\}$

[] $(A - B) \cap B = \{x | x \in A \vee x \in B\}$

[] $(A - B) \cap B = \{\}$

[] $(A - B) \cap B = \{x | (x \in A \wedge x \notin B) \wedge x \in B\}$

6. (15 points) Let A , B and C be three finite sets. Disprove with a counterexample:

$$|A \cup (B \cap C)| = |A| + |B \cap C|$$

7. (20 points) Let A and B be two sets. Prove the following in two ways: using an **element-chasing proof**, and using **set identities**. Justify each step in both proofs.

$$A \cup B = A \cap B \implies A = B$$



Credit: Image LLM-generated